

Alan Clark - Elements of Abstract Algebra

Chapter 2, Exercise 26 α

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26 α . A semigroup is a set S with a product which associates to each ordered pair (a, b) of elements of S an element $ab \in S$ in such a way that $(ab)c = a(bc)$ for any elements $a, b, c \in S$.

Show that the set of all mappings from a given set X to itself forms a semigroup in which the product is composition of mappings.

Show that the set of all one-to-one correspondences of X with itself forms a group under composition.

Let $X^X := \{f : X \rightarrow X\}$. We aim to show that $(X^X, \circ)$ is a semigroup.

Let $f, g \in X^X$. Then $(f \circ g) : X \rightarrow X$, that is, $(f \circ g) \in X^X$. So X^X is closed under composition.

Let $f, g, h \in X^X$ and let $x \in X$. We have

$$\begin{aligned}(f \circ (g \circ h))(x) &= (f \circ (g \circ h)(x)) \\ &= (f \circ (g(h(x)))) \\ &= f(g(h(x))) \\ &= (f \circ g) \circ h(x) \\ &= ((f \circ g) \circ h)(x)\end{aligned}$$

Thus, the composition is associative and, therefore $(X^X, \circ)$ is a semigroup.

Let $\overline{X}^X := \{f : X \rightarrow X \mid f \text{ is bijective}\}$. We have to show that $(\overline{X}^X, \circ)$ is a group.

Let $f, g \in \overline{X}^X$. Then $(f \circ g) : X \rightarrow X$. Moreover $(f \circ g)$ is bijective. So $\overline{X}^X$ is closed under composition.

- (associativity) The same proof used for X^X works for $\overline{X}^X$.

- (identity) Consider the identity function, $i : X \rightarrow X$. Clearly, i is bijective, so $i \in \overline{X}^X$. Let $f \in \overline{X}^X$. By definition,

$$f \circ i = f = i \circ f$$

- (inverse) Let $f \in \overline{X}^X$. Then f is bijective.

Now construct $g : X \rightarrow X$ such that $g(y) = x$ iff $f(x) = y$. For g to be a function it must be the case that

$$\begin{aligned} \forall y \in X \exists! x \in X \quad g(y) = x \\ \Leftrightarrow \\ \forall y \in X \exists! x \in X \quad f(x) = y \end{aligned}$$

but the last condition is true because f is bijective.

Let's verify that $g \in \overline{X}^X$, that is, that g is bijective.

Suppose that $g(y_1) = g(y_2)$ for some y_1, y_2 . Let's denote $x_1 := g(y_1)$ and $x_2 := g(y_2)$. Thus, by definition of g , we have that $f(x_1) = y_1$ and $f(x_2) = y_2$. Since $x_1 = x_2$, of course, $y_1 = y_2$, that is, g is injective.

Let $x \in X$, then there exists a $y \in X$ such that $f(x) = y$, that is, $g(y) = x$. Thus, g is onto.

Therefore, g is bijective, that is, $g \in \overline{X}^X$.

Let's verify that g , defined that way, is indeed the inverse of f ,

$$\forall x \in X, \quad (g \circ f)(x) = g(f(x)) = x$$

and

$$\forall y \in X, \quad (f \circ g)(y) = f(g(y)) = y.$$

Thus, $g = f^{-1}$, as desired.

Therefore, $(\overline{X}^X, \circ)$ is a group.